

입자물리개론(이) (PHY409-01)

학기	2026년 2학기
담당교수	DEVECIOGLU DENIZ OLGU
이수구분	교선평수
학점	3.0
강의시간	화14:30-16:00(E7 - 229), 목14:30-16:00(E7 - 229)
강의실	
소속	기초학부

교과개요

This course surveys modern perspectives on the fundamental constituents of matter and their interactions. Utilizing Quantum Field Theory (QFT) as a foundational framework, the course emphasizes practical results and conceptual understanding. While this is not a full QFT course, we will use its results to understand particle physics, meaning familiarity with classical mechanics and quantum mechanics is sufficient. Students will analyze the structure and content of the Standard Model, explore its cross-disciplinary connections to other fields of physics, and examine the theoretical motivations for pushing beyond its boundaries.

교과목표

The first part of the semester we will focus on the formal structure of the Standard Model. We will build up necessary mathematical tools for treatment of symmetries, special relativity, relativistic Lagrangians and equations of motion for scalar, spinor and vector fields, the principle of local gauge invariance and the Higgs mechanism for mass generation. Equipped with these tools, in the second part of the course will focus more on computational aspects of QFT. Such as a review of perturbation theory and how it is applied to relativistic field theories. Here we will use Feynman diagrams and reproduce some of the well known predictions of Standard Model.

주차별 강의계획

- 1주차: Introduction to Course, Groups and Representations
- 2주차: Group metrics, Special Relativity
- 3주차: Energy and Momentum in Special Relativity
- 4주차: Lie groups, Lie algebras, Spinors
- 5주차: Lagrangians, Action Principles, Degrees of Freedom
- 6주차: Dirac Equation, Helicity and Weyl Spinors
- 7주차: Interactions via Local Gauge Symmetry, QCD as an SU(3) Gauge Theory
- 8주차: Electroweak gauge theory
- 9주차: Higgs Mechanism
- 10주차: Decays and Scattering
- 11주차: Feynman Rules
- 12주차: More on Feynman Rules
- 13주차: QCD and calculations 1
- 14주차: QCD and calculations 2
- 15주차: Weak Interactions
- 16주차: Finals

학습윤리

You are expected to demonstrate academic honesty in all aspects of this course. Academic dishonesty includes, but is not limited to: violating clearly stated rules for taking an exam or completing an assignment; plagiarism (including material from sources without a citation and quotation marks around any borrowed words); claiming another's work or a modification of another's work as one's own; buying or attempting to buy papers or projects for a course; fabricating information or citations.

교과정책(방침)

Attendance is mandatory and will be formally graded. Assignments and exams must be submitted on time. Late submissions will result in point deductions that may affect your final grade. If you anticipate a conflict with a due date or encounter difficulties with an assignment, you must plan ahead and notify the instructor in advance. This will allow us to discuss the situation and, if appropriate, make alternative arrangements.